

CE480 - HW5

Preliminary Small Commercial Harbor Layout and Breakwater Design

Group 3

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1 Executive Summary

This report presents a preliminary harbor-layout and breakwater-design calculation for a small commercial harbor at the same site used in the previous CE480 homeworks. The design uses a maximum vessel length of 80 m, maximum draft of 5.5 m, and maximum beam of 14 m. A quay wall length of 150 m is adopted based on similar small commercial harbor scales.

The calculated navigation depth is **7.1 m**. The estimated bathymetry slope from the Google Earth Pro profile is **0.0196 m/m**, approximately **1:51**. The required depth is reached at approximately **524 m** from the shore. Therefore, the selected perpendicular breakwater extension is **575 m**, giving a local water depth of **8.1 m** at the breakwater head/toe. The parallel arm is estimated as **400 m**, so the total preliminary breakwater length is **975 m**.

From HW4, the 10-year and 50-year deep-water design wave heights are **5.59 m** and **8.04 m**, respectively. Using the course-style Goda nearshore design-wave check at the selected breakwater depth, the corresponding design wave heights in front of the breakwater are **5.13 m** and **5.38 m**. The governing condition is depth-limited breaking.

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2 Problem Statement and Methodology

The homework requires the harbor dimensions and wave-design checks listed below:

1. Select representative maximum and mean ship sizes for the selected harbor type.
2. Estimate quay wall or pier length from similar harbor scale.
3. Calculate the turning basin.
4. Determine the breakwater extension perpendicular and parallel to shore.
5. Determine the necessary water depth using a navigation-channel depth formula.
6. Check whether the breakwater reaches the required depth; if not, propose a navigation channel.
7. Use HW4 10-year and 50-year deep-water wave heights and calculate the design wave height in front of the breakwater using Goda's formula with wave steepness $s_0 = 0.04$.

The workflow is summarized as follows:

Ship size $\rightarrow$ quay and basin dimensions $\rightarrow$ navigation depth $\rightarrow$ breakwater reach $\rightarrow$ Goda design wave

3 Input Data and Assumptions

3.1 Site and wave inputs

The site and wave inputs are taken from the previous CE480 homework workflow. The critical wave direction is the direction with the largest effective fetch. The fetch and wave-return values are treated as fixed inputs for this HW5 preliminary design.

3.2 External design basis

The harbor layout calculations are preliminary, but the adopted approach follows common harbor-planning guidance. PIANC guidance for harbor approach channels is used as the main external basis for approach-channel and maneuvering-area assumptions, while Thoresen's port design handbook is used as a general port-planning reference for quay and harbor layout concepts [1, 2]. The wave and breakwater design explanations are supported by the USACE Coastal Engineering Manual and Goda's random-sea design reference [3, 4]. Bathymetry was estimated from a Google Earth Pro elevation profile, and GEBCO is included as an external reference for global bathymetry data products that should be used for checking or improving the preliminary profile [7, 6].

Table 1: Main input data used in HW5.

Input	Value	Unit
Site latitude	41.934003	deg
Site longitude	28.069026	deg
Harbor type	Small commercial harbor	-
Critical wave direction	67.5	deg
Effective fetch	1074.12	km
Deep-water wave steepness, s_0	0.04	-
Bathymetry point 1	457.2 m, depth 5.79 m	-
Bathymetry point 2	2027.8 m, depth 36.58 m	-
Estimated seabed slope	0.0196	m/m
Approximate seabed slope	1:51	-

3.3 Design vessel and quay assumptions

For a small commercial harbor, the design vessel was selected as a small general-cargo/coaster-type vessel. Since no specific vessel fleet was assigned in the homework statement, the values below are preliminary assumptions and should be replaced if the instructor requires a specific vessel type.

Table 2: Design vessel, quay, and harbor-basin assumptions.

Item	Value	Unit
Mean ship length	50	m
Maximum/design ship length, L_{max}	80	m
Mean draft	4.0	m
Maximum/design draft, T_d	5.5	m
Mean beam	9	m
Maximum/design beam, B_{max}	14	m
Quay wall length	150	m
Entrance width	100	m

4 Turning Basin, Channel Width, and Quay Length

The turning basin diameter is taken as twice the maximum design vessel length. This is a conservative preliminary choice for a small commercial harbor and is consistent with commonly used harbor-maneuvering guidance where turning diameter is linked to the design vessel length [1, 2]:

$$D_T = 2L_{max} \quad (1)$$

$$D_T = 2(80) = 160 \text{ m} \quad (2)$$

The one-way preliminary navigation channel width is estimated as:

$$B_c = 5B_{max} \quad (3)$$

$$B_c = 5(14) = 70 \text{ m} \quad (4)$$

The adopted quay wall length is 150 m. This provides berthing space for approximately one small commercial vessel with operational allowance, or a shorter combination of small service vessels.

5 Navigation Depth Calculation

The required navigation depth is computed by adding the design draft, under-keel clearance, wave allowance, dredging tolerance, and sedimentation allowance. This follows the general navigation-channel concept that safe water depth is the vessel draft plus allowances for operational and environmental uncertainty [1]:

$$D_{req} = T_d + UKC + A_w + A_d + A_s \quad (5)$$

where T_d is the design vessel draft, UKC is under-keel clearance, A_w is wave allowance, A_d is dredging tolerance, and A_s is sedimentation allowance.

The under-keel clearance is assumed as 10% of the design draft:

$$UKC = 0.10T_d = 0.10(5.5) = 0.55 \text{ m} \quad (6)$$

Therefore,

$$D_{req} = 5.5 + 0.55 + 0.50 + 0.30 + 0.20 = 7.05 \text{ m} \quad (7)$$

After rounding upward, the adopted required navigation depth is:

$$D_{req} = 7.1 \text{ m} \quad (8)$$

Table 3: Navigation depth calculation.

Component	Value (m)
Design draft	5.50
Under-keel clearance (10% draft)	0.55
Wave allowance	0.50
Dredging tolerance	0.30
Sedimentation allowance	0.20
Required depth before rounding	7.05
Required design depth	7.10

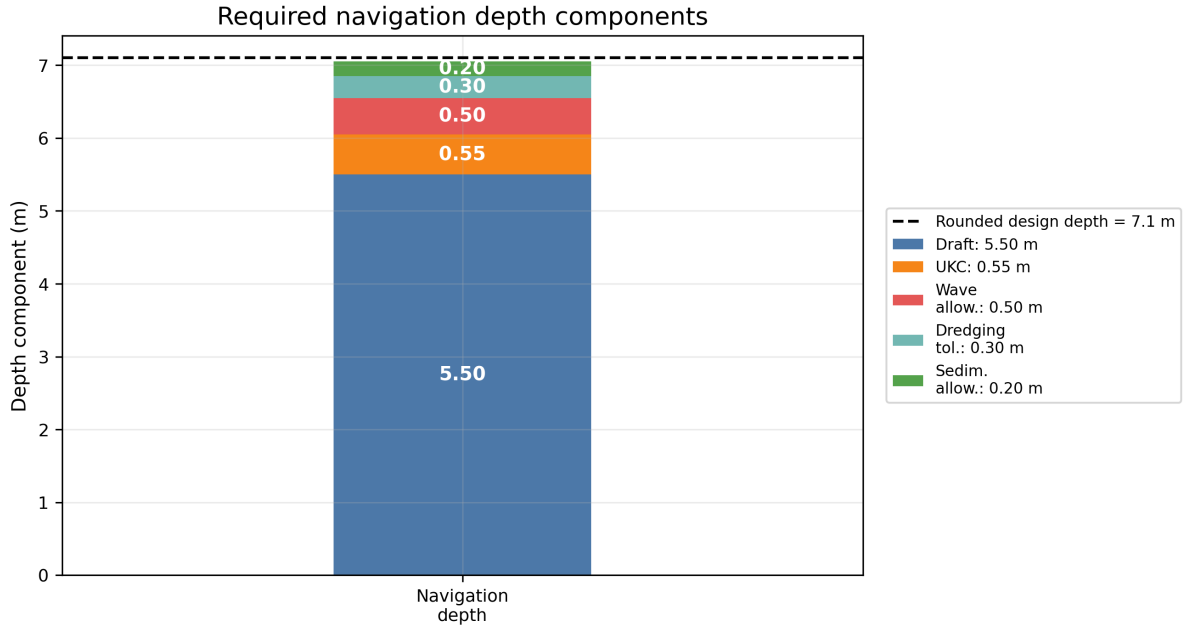


Figure 1: Components of the required navigation depth.

6 Bathymetry, Required Depth, and Breakwater Reach

The Google Earth Pro profile points were converted from English units to SI units. Google Earth Pro can display distance and elevation along a saved path, but the result should be treated as an approximate screening profile rather than a hydrographic survey [7]. A linear bathymetry model was fitted between the two points:

$$h(x) = h_0 + mx \quad (9)$$

where x is offshore distance and h is water depth. The estimated slope is:

$$m = \frac{36.58 - 5.79}{2027.8 - 457.2} = 0.0196 \approx 1 : 51 \quad (10)$$

The distance where the required depth is reached is:

$$x_{req} = \frac{D_{req} - h_0}{m} = 524 \text{ m} \quad (11)$$

To ensure the breakwater reaches slightly deeper water than the minimum navigation requirement, an additional 50 m is added and the result is rounded up:

$$L_{BW,\perp} = \text{round up}(x_{req} + 50) = 575 \text{ m} \quad (12)$$

The resulting depth at the breakwater head/toe is approximately:

$$h_{toe} = 8.1 \text{ m} \quad (13)$$

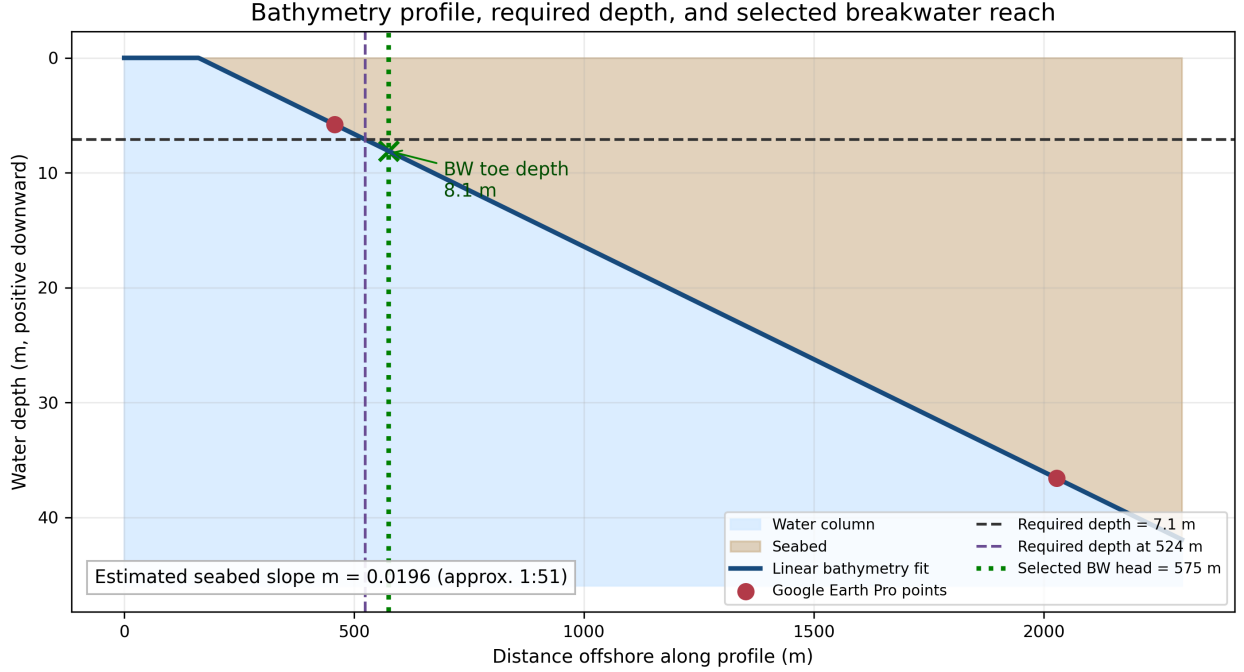


Figure 2: Bathymetry profile, required navigation depth, and selected breakwater head distance.

7 Breakwater Layout

The preliminary breakwater layout is composed of two parts:

1. A perpendicular arm extending from shore until the required water depth is exceeded.
2. A parallel arm protecting the quay and turning basin from the critical wave direction.

The parallel breakwater length is estimated from the quay length, turning basin diameter, and an additional protection margin:

$$L_{BW,\parallel} = L_q + D_T + M_p \quad (14)$$

$$L_{BW,\parallel} = 150 + 160 + 80 = 390 \text{ m} \approx 400 \text{ m} \quad (15)$$

Thus, the total preliminary breakwater length is:

$$L_{BW,total} = 575 + 400 = 975 \text{ m} \quad (16)$$

Table 4: Summary of harbor layout dimensions.

Item	Value	Unit
Turning basin diameter	160	m
Channel width	70	m
Required navigation depth	7.1	m
Distance to required depth	524	m from shoreline
Perpendicular breakwater length	575	m
Depth at breakwater head/toe	8.1	m
Parallel breakwater length	400	m
Total breakwater length	975	m

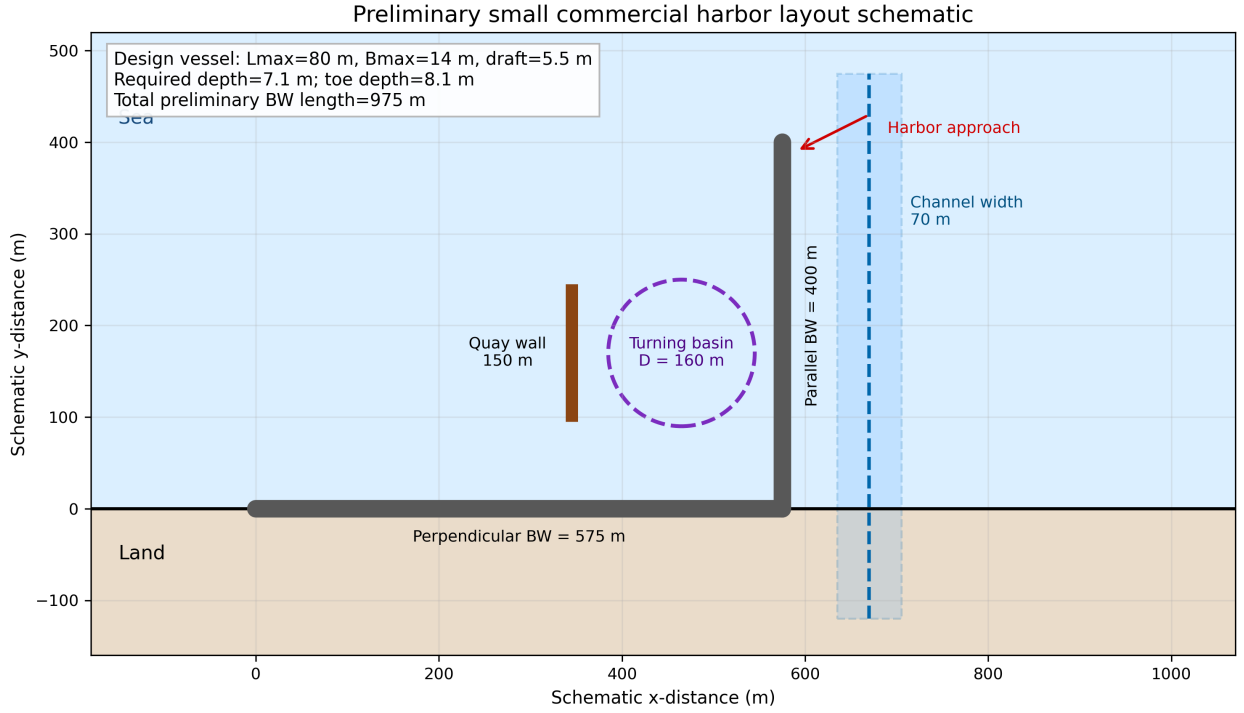


Figure 3: Preliminary harbor layout schematic. The figure is a conceptual layout, not a georeferenced engineering drawing.

8 Deep-Water Design Waves from HW4

The deep-water return wave heights were obtained using the Gumbel extreme-value method. Extreme-wave analysis is a standard coastal engineering step for estimating return-period design conditions before transferring waves to the structure toe [3]. The plotting position used in the previous HW4 code is the Gringorten form:

$$P = 1 - \frac{m - C_1}{N + C_2}, \quad C_1 = 0.44, \quad C_2 = 0.12 \quad (17)$$

where m is rank and N is the number of annual maxima. The Gumbel reduced variate is:

$$y = -\ln[-\ln(P)] \quad (18)$$

The fitted relation is:

$$H_s = ay + b = 1.4849y + 2.2476, \quad R^2 = 0.969 \quad (19)$$

For a return period T_R , the exceedance probability is $Q = 1/T_R$ and the non-exceedance probability is $P = 1 - Q$.

Table 5: Deep-water return-period wave heights used in HW5.

T_R (yr)	P	y	H_0 (m)
10	0.900	2.250	5.59
50	0.980	3.902	8.04
100	0.990	4.600	9.08

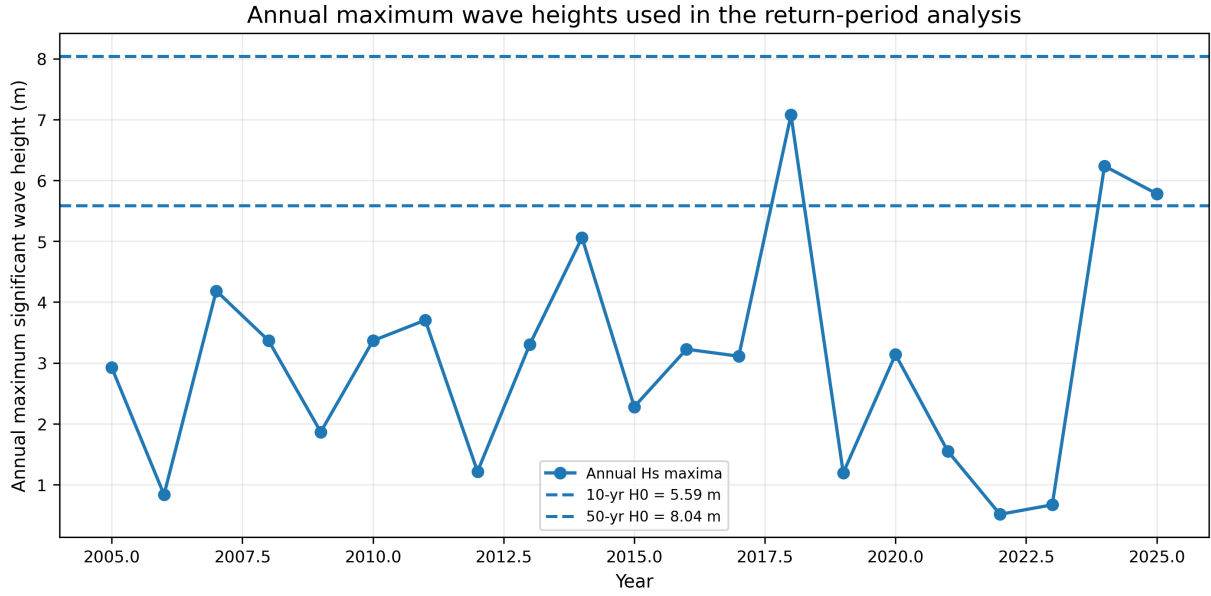


Figure 4: Annual maximum significant wave heights and 10-year/50-year return wave levels.

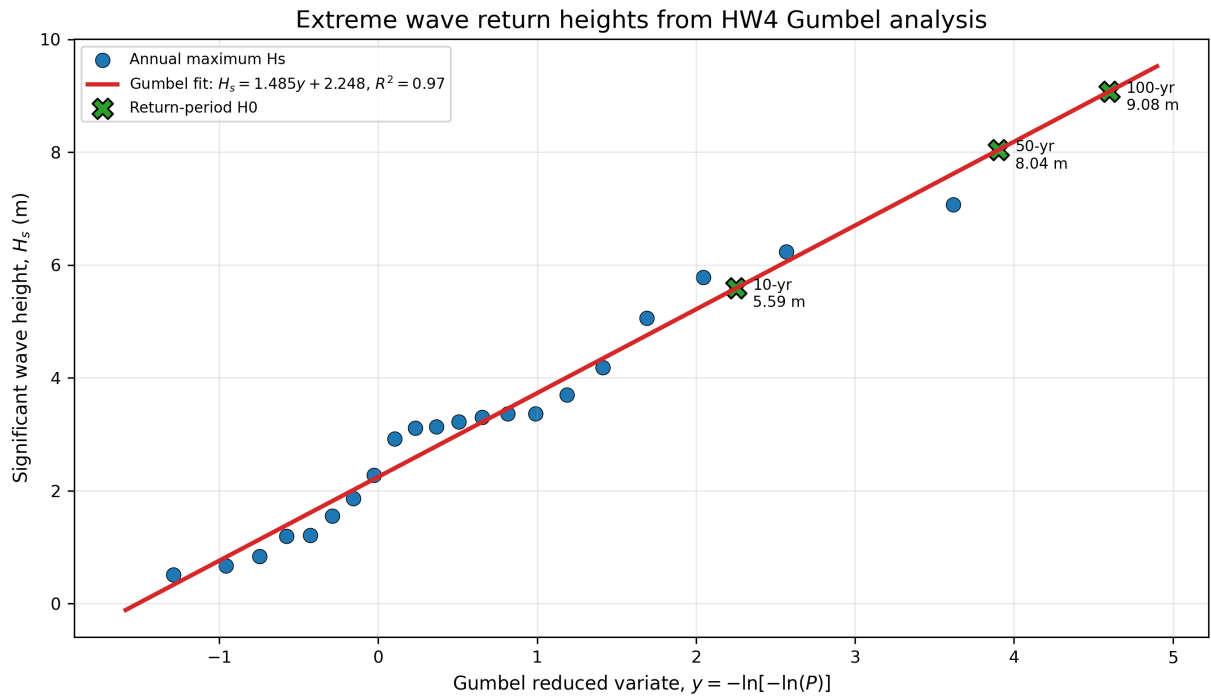


Figure 5: Gumbel fit and return-period wave heights.

9 Goda Design Wave Height at the Breakwater

The design wave height in front of the breakwater is evaluated at the local depth $h = 8.1$ m. The wave steepness is given by the homework statement as:

$$s_0 = \frac{H_0}{L_0} = 0.04 \quad (20)$$

Therefore, for each return-period wave:

$$L_0 = \frac{H_0}{s_0} \quad (21)$$

and the wave period is obtained from the deep-water dispersion relation:

$$L_0 = \frac{gT^2}{2\pi} \Rightarrow T = \sqrt{\frac{2\pi L_0}{g}} \quad (22)$$

The course-style Goda nearshore wave-height check is implemented as:

$$H_d = \min(K_s H_0, \beta_0 H_0 + \beta_1 h, \beta_{max} H_0) \quad (23)$$

where K_s is the shoaling coefficient, h is the local water depth, and the β coefficients depend on wave steepness and seabed slope. The coefficient set used in the calculation is:

$$\beta_0 = 0.028 s_0^{-0.38} \exp(20m^{1.5}) \quad (24)$$

$$\beta_1 = 0.52 \exp(4.2m) \quad (25)$$

$$\beta_{max} = \max[0.92, 0.32 s_0^{-0.29} \exp(2.4m)] \quad (26)$$

where $m = 0.0196$ is the seabed slope. The selected design wave is the smallest of the three candidate terms because it represents the limiting transformation/breaking condition at the breakwater location.

Table 6: Goda-type design wave calculation results.

T_R (yr)	H_0 (m)	T (s)	h (m)	H_d (m)	Governing condition
10	5.59	9.46	8.10	5.13	Depth-limited breaking
50	8.04	11.35	8.10	5.38	Depth-limited breaking
100	9.08	12.06	8.10	5.49	Depth-limited breaking

Table 7: Candidate terms in Goda design wave calculation.

T_R (yr)	$K_s H_0$ (m)	$\beta_0 H_0 + \beta_1 h$ (m)	$\beta_{max} H_0$ (m)	Selected H_d (m)
10	5.58	5.13	5.14	5.13
50	8.55	5.38	7.40	5.38
100	9.87	5.49	8.35	5.49

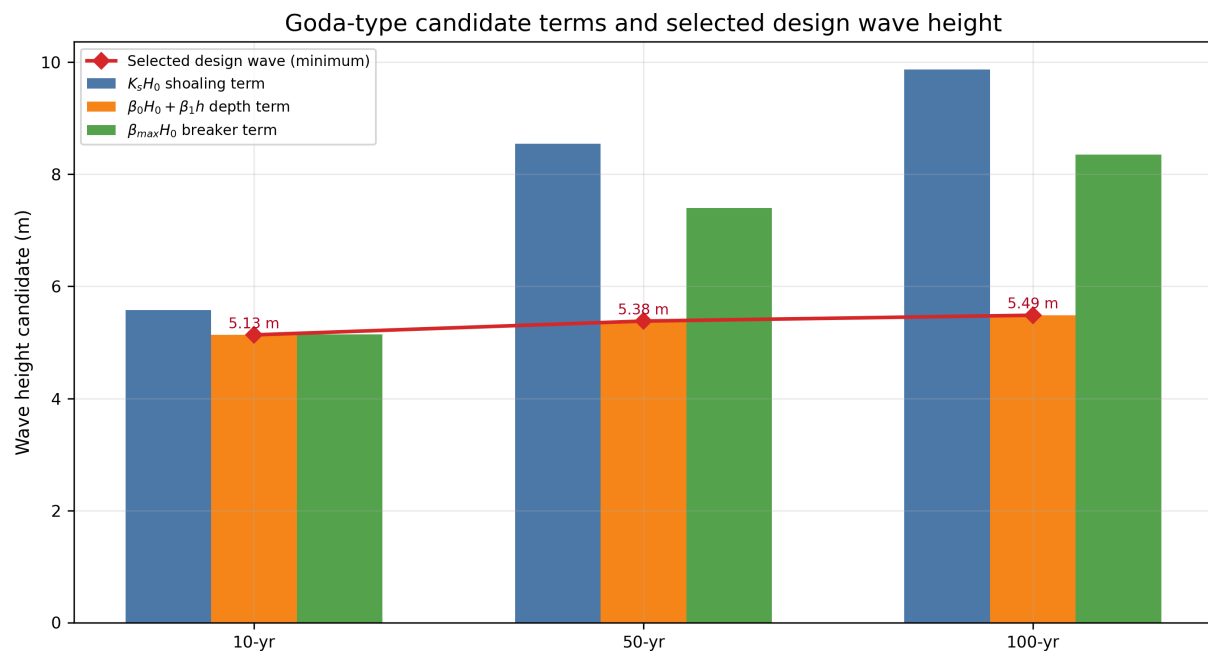


Figure 6: Goda-type candidate terms and selected design wave height.

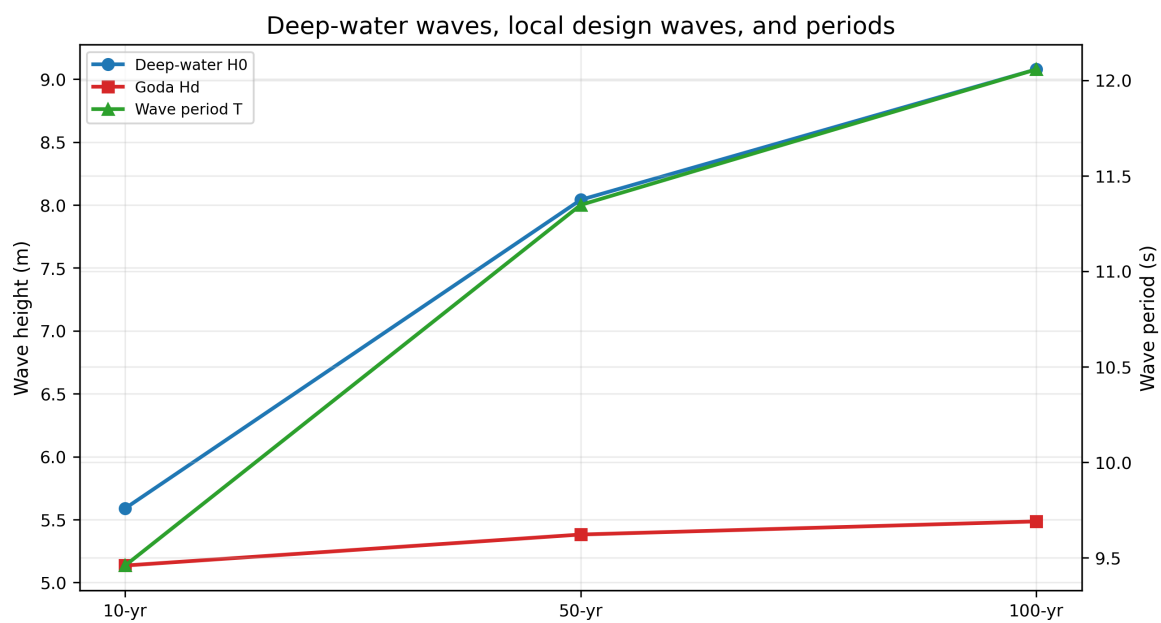


Figure 7: Comparison of deep-water wave height, local design wave height, and wave period.

10 Final Design Summary

Table 8: Final preliminary design values.

Design item	Adopted/calculated value
Harbor type	Small commercial harbor
Maximum/design vessel length	80 m
Maximum/design draft	5.5 m
Maximum/design beam	14 m
Quay wall length	150 m
Turning basin diameter	160 m
Navigation channel width	70 m
Required navigation depth	7.1 m
Perpendicular breakwater length	575 m
Parallel breakwater length	400 m
Total preliminary breakwater length	975 m
10-year deep-water wave height, $H_{0,10}$	5.59 m
50-year deep-water wave height, $H_{0,50}$	8.04 m
10-year Goda design wave, $H_{d,10}$	5.13 m
50-year Goda design wave, $H_{d,50}$	5.38 m
Governing wave transformation condition	Depth-limited breaking

11 Conclusions

1. A small commercial harbor layout was developed using a design vessel length of 80 m and design draft of 5.5 m.
2. The required navigation depth is 7.1 m. Based on the linear bathymetry approximation, this depth occurs approximately 524 m offshore.
3. The selected perpendicular breakwater length of 575 m reaches a local depth of about 8.1 m, so the breakwater reaches the required navigation depth. A navigation channel may still be marked or dredged for safe access, but the preliminary layout satisfies the depth requirement.
4. The preliminary parallel breakwater length is 400 m, giving a total preliminary breakwater length of 975 m.
5. The Goda-type design wave heights at the breakwater are 5.13 m for the 10-year return period and 5.38 m for the 50-year return period. For both cases, the governing condition is depth-limited breaking.

12 External References

- [1] PIANC. (2014). *Harbour Approach Channels Design Guidelines*. Report No. 121, Maritime Navigation Commission (MarCom), The World Association for Waterborne Transport Infrastructure.
- [2] Thoresen, C. A. (2014). *Port Designer's Handbook: Recommendations and Guidelines*. 3rd edition, ICE Publishing / Thomas Telford.

- [3] U.S. Army Corps of Engineers. (2002). *Coastal Engineering Manual*, EM 1110-2-1100, Parts II and VI. U.S. Army Corps of Engineers, Washington, DC. Available through the official USACE Publications repository.
- [4] Goda, Y. (2010). *Random Seas and Design of Maritime Structures*. 3rd edition, World Scientific.
- [5] CIRIA, CUR, and CETMEF. (2007). *The Rock Manual: The Use of Rock in Hydraulic Engineering*. 2nd edition, CIRIA C683, London.
- [6] GEBCO Compilation Group. (2024). *GEBCO Gridded Bathymetry Data*. General Bathymetric Chart of the Oceans. <https://www.gebco.net/data-products/gridded-bathymetry-data>
- [7] Google. (n.d.). *Measure distance and elevation - Google Earth Help*. <https://support.google.com/earth/answer/148134>